COMPUTER NETWORKS

Lecture-06

Recap Of Lecture- 05

- The OSI Model
- Layered task
- Peer-to-Peer communication
- OSI Layers

Comparison and Contrast between the OSI and TCP/IP Model

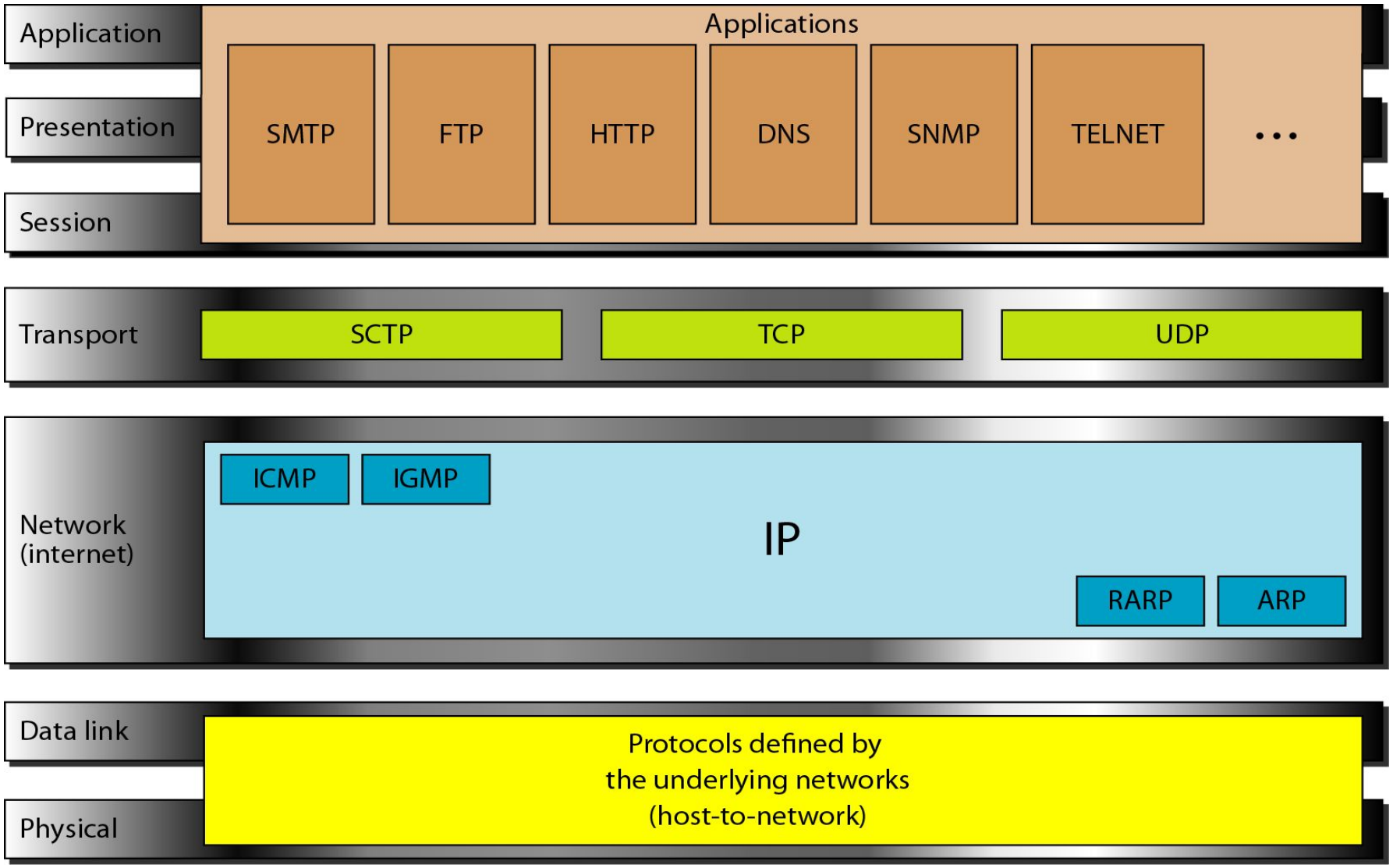
TCP/IP PROTOCOL SUITE

- The **Internet protocol suite** is the computer networking model and set of communications protocols used on the Internet and similar computer networks.
- It is commonly known as TCP/IP, because its most important protocols, the Transmission Control Protocol (TCP) and the Internet Protocol (IP), were the first networking protocols defined in this standard.
- It was originally known as the **DoD model**, because the development of the networking model was funded by **DARPA**, US **Department of Defense**
- TCP/IP provides end-to-end connectivity specifying how data should be packetized, addressed, transmitted, routed and received at the destination.
- The TCP/IP model and related protocol models are maintained by the **Internet Engineering Task Force** (IETF).

TCP/IP PROTOCOL SUITE

- The layers in the **TCP/IP protocol suite** do not exactly match those in the OSI model.
- The original TCP/IP protocol suite was defined as having **four layers**:
  1. Host-to-network (subnet layer/Link Layer/Network Interface Layer),
  2. Internet,
  3. Transport, and
  4. Application.
- However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of **five layers**:
  1. Physical,
  2. Data link,
  3. Network,
  4. Transport, and
  5. Application.

TCP/IP and OSI model



Upper Layers

OSI	TCP / IP
Application (Layer7)	Application
Presentation (Layer6)	
Session (Layer 5)	

Session Layer

- The Session layer permits two parties to hold ongoing communications called a session across a network.
- Not found in TCP/IP model
- In TCP/IP, its characteristics are provided by the TCP protocol. (Transport Layer)

Presentation Layer

- The Presentation Layer handles data format information for networked communications.
- This is done by converting data into a generic format that could be understood by both sides.
- Not found in TCP/IP model
- In TCP/IP, this function is provided by the Application Layer.
- e.g.
  - External Data Representation Standard (XDR)
  - Multipurpose Internet Mail Extensions (MIME)

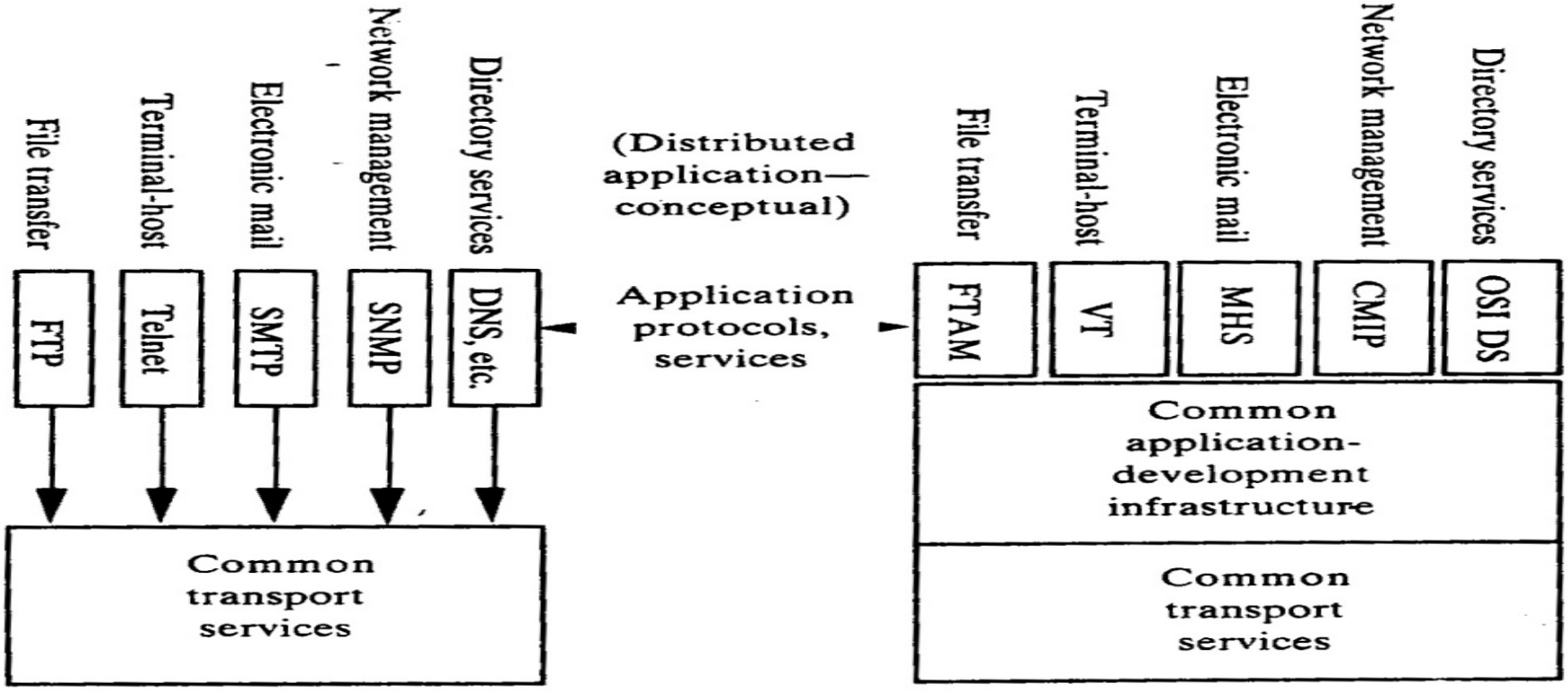
Application Layer

- It provides a set of interfaces for applications to obtain access to networked services as well as access to the kinds of network services that support applications directly.

- Although the notion of an application process is common to both, their approaches to constructing application entities is different.

Approaches use in constructing application entities

- The diagram below provides an overall view on the methods use by both the OSIThe diagram below provides an overall view on the methods use by both the OSI and TCP/IP model.



Transport Layer

OSI	TCP / IP
Transport (Layer 4)	Transport (TCP/UDP)

- The functionality of the transport layer is to provide “transparent transfer of data from a source end open system to a destination end open system”.

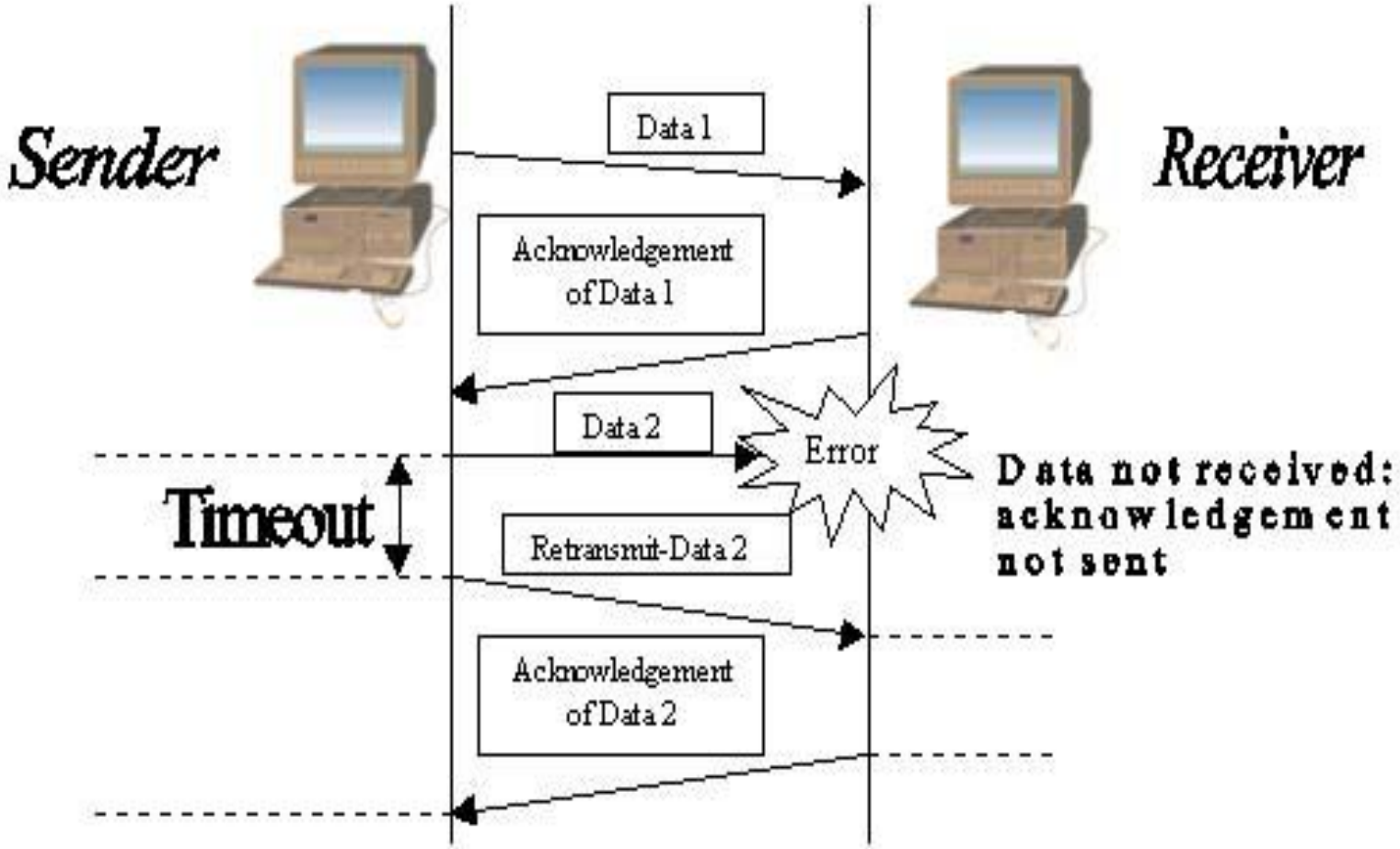
Transport Layer

- Transport is responsible for;
  - Creating and maintaining the basic end-to-end connection between communicating open systems,
  - Ensuring that the bits delivered to the receiver are the same as the bits transmitted by the sender;
  - In the same order and without modification, loss or duplication

Transport Layer

- Segmentation and reassembly
- Capability to acknowledge and retransmit if no acknowledgement is received
- Mark packets with sequencing information so that the destination system can properly order the packets if they're received out-of-sequence
- Transport protocols provide the capability for multiple application processes to access the network by using individual local addresses to determine the destination process for each data stream

Transport Layer



TCP/IP Transport Layer

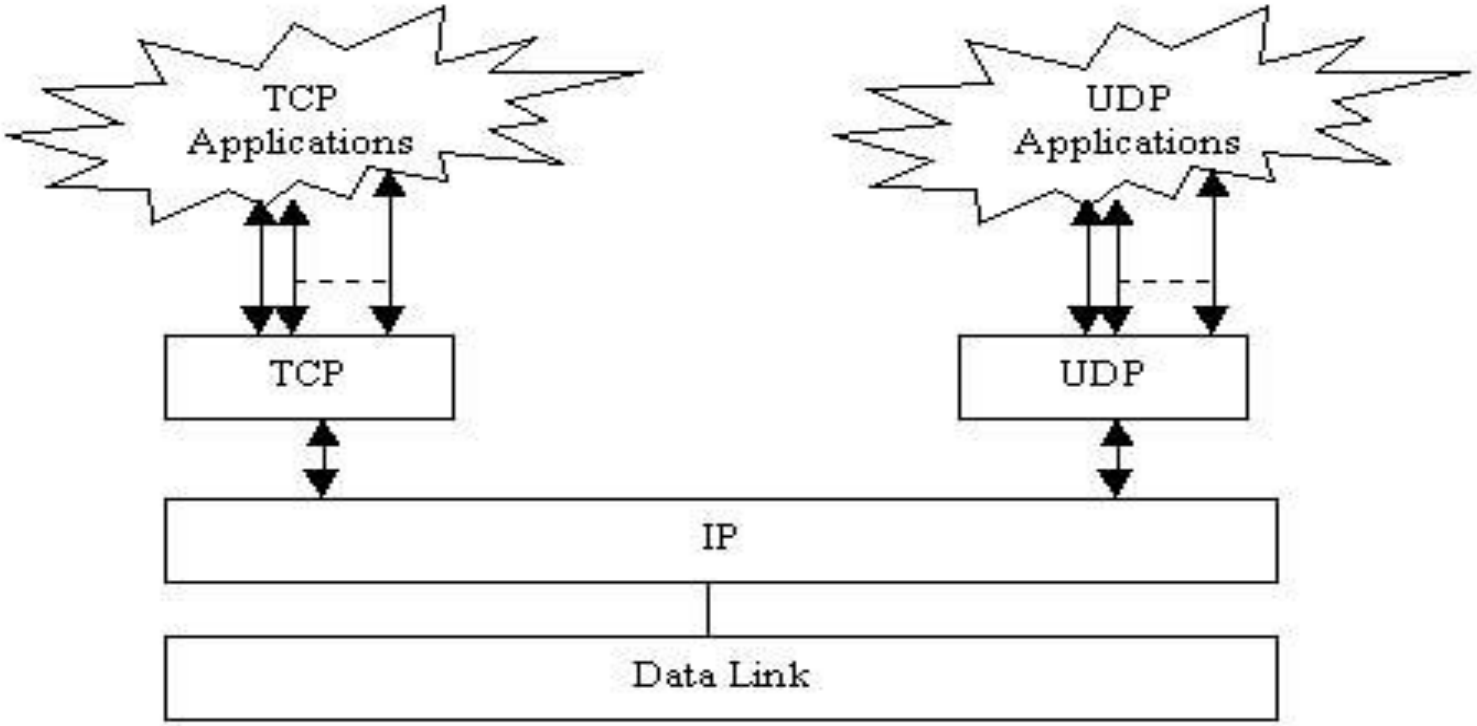
- Defines two standard transport protocols:

1. **TCP**
   - implements a reliable data-stream protocol
   - connection oriented

2. **UDP**
   - implements an unreliable data-stream
   - connectionless
   - useful in many applications where data needs to be broadcasted or multi-casted
   - **"Primary difference is that UDP does not necessarily provide reliable data transmission"**

TCP/IP Transport Layer

- Many programs will use a separate TCP connection as well as a UDP connection



TCP/IP Transport Layer

- TCP and UDP introduce the concept of ports

- Common ports and the services that run on them:
  - FTP 21 and 20
  - telnet 23
  - SMTP 25
  - http 80
  - POP3 110

TCP/IP Transport Layer

- By specifying ports and including port numbers with TCP/UDP data, multiplexing is achieved
- Multiplexing allows multiple network connections to take place simultaneously
- The port numbers, along with the source and destination addresses for the data, determine a socket

Internet Layer

OSI	TCP / IP
Network (Layer 3)	Internet Layer

Internet Layer

- At this Layer, TCP/IP supports the Internetworking Protocol.

- IP, in turn, uses four supporting protocols:
  1. ARP,
  2. RARP,
  3. ICMP, and
  4. IGMP.

Data link / Physical vs. Subnet

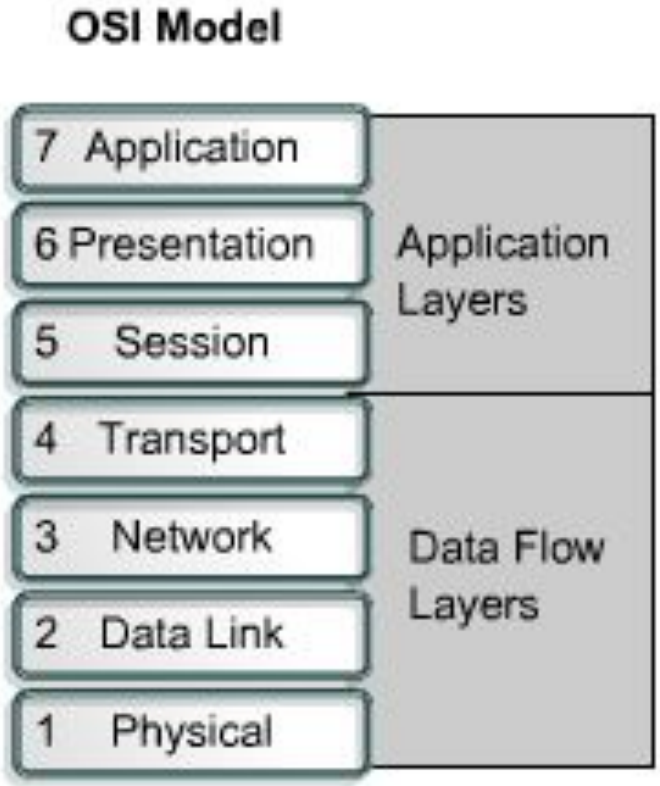
OSI	TCP / IP
Data Link (Layer 2)	Subnet
Physical (Layer 1)	

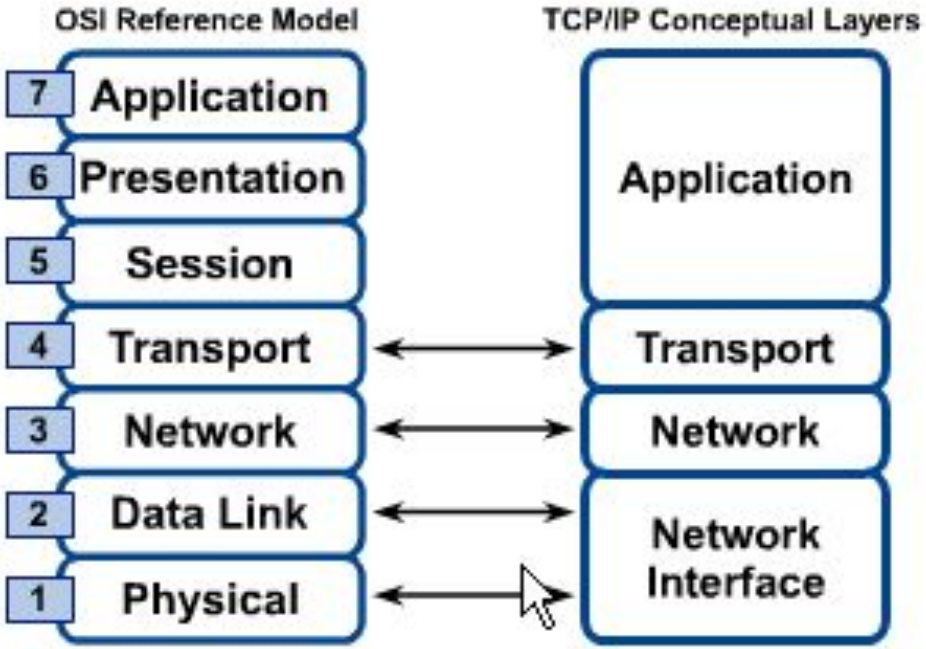
- **Data link layer**
  - It provides Flow control, and detects and possibly corrects errors which may occur".
  - In another words, the Data Link Layer transforms a stream of raw bits (0s and 1s) from the physical into a data frame and provides an error-free transfer from one node to another, allowing the layers above it to assume virtually error-free transmission.

Data link / Physical vs. Subnet

- **Physical layer**
  - The function of the Physical Layer is to provide "**mechanical, electrical, functional, and procedural** means to activate a physical connection for bit transmission".
  - Basically, this means that the typical role of the physical layer is to transform bits in a computer system into electromagnetic (or equivalent) signals for a particular transmission medium (wire, fiber, ether, etc.)

OSI & TCP/IP Models





TCP/IP Model

Application Layer
Application programs using the network

Transport Layer (TCP/UDP)
Management of end-to-end message transmission, error detection and error correction

Network Layer (IP)
Handling of datagrams : routing and congestion

Data Link Layer
Management of cost effective and reliable data delivery, access to physical networks

Physical Layer
Physical Media

Suggested Reading

- Section
  - 2.4,
    - "Data Communications and Networking" 4th Edition by Behrouz A. Forouzan